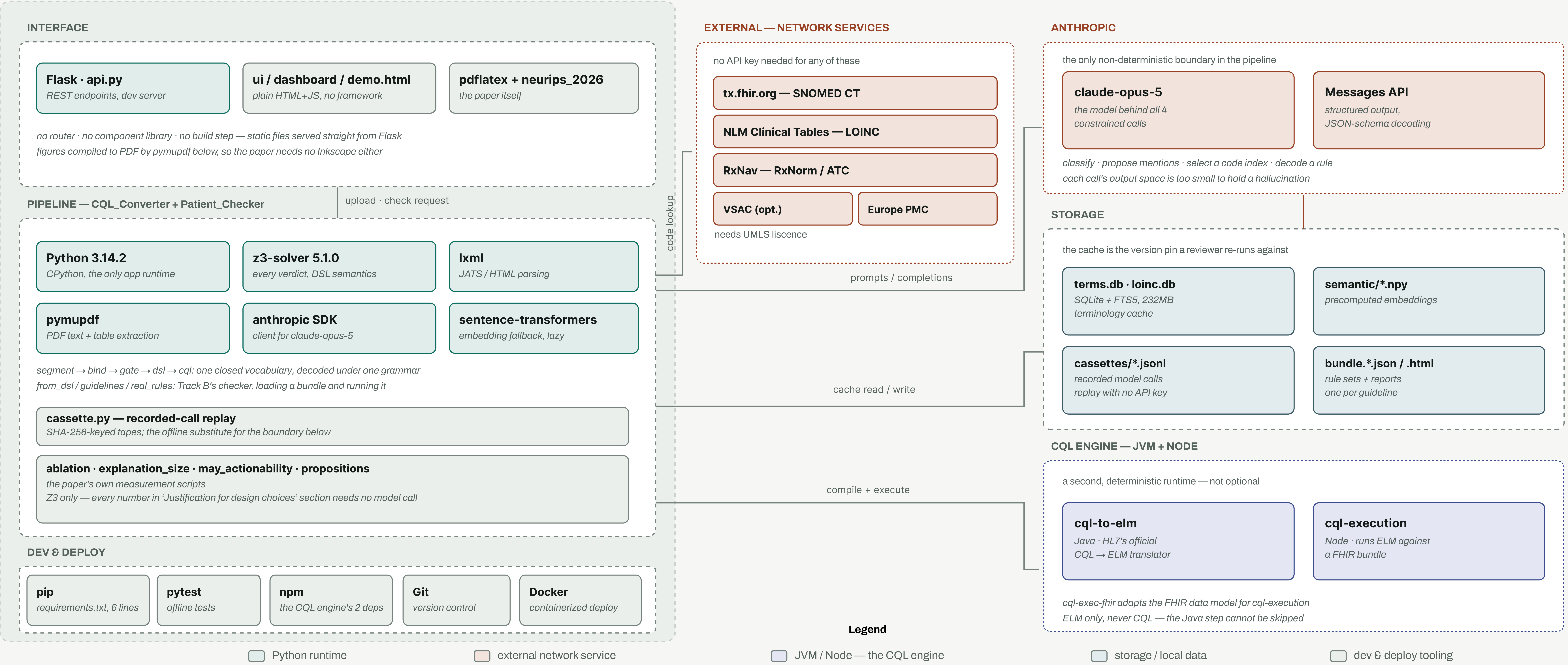
GUIDELINEFORGE



INTERFACE

Flask · api.py
REST endpoints, dev server

ui / dashboard / demo.html
plain HTML+JS, no framework

pdflatex + neurips_2026
the paper itself

no router · no component library · no build step — static files served straight from Flask
figures compiled to PDF by pymupdf below, so the paper needs no Inkscape either

upload · check request

PIPELINE — CQL_Converter + Patient_Checker

Python 3.14.2
CPython, the only app runtime

z3-solver 5.1.0
every verdict, DSL semantics

lxml
JATS / HTML parsing

pymupdf
PDF text + table extraction

anthropic SDK
client for claude-opus-5

sentence-transformers
embedding fallback, lazy

segment → bind → gate → dsl → cql: one closed vocabulary, decoded under one grammar
from_dsl / guidelines / real_rules: Track B's checker, loading a bundle and running it

cassette.py — recorded-call replay
SHA-256-keyed tapes; the offline substitute for the boundary below

ablation · explanation_size · may_actionability · propositions
the paper's own measurement scripts
Z3 only — every number in 'Justification for design choices' section needs no model call

DEV & DEPLOY

pip
requirements.txt, 6 lines

pytest
offline tests

npm
the CQL engine's 2 deps

Git
version control

Docker
containerized deploy

code lookup

EXTERNAL — NETWORK SERVICES

no API key needed for any of these

tx.fhir.org — SNOMED CT

NLM Clinical Tables — LOINC

RxNav — RxNorm / ATC

VSAC (opt.)

Europe PMC

needs UMLS liscence

prompts / completions

ANTHROPIC

the only non-deterministic boundary in the pipeline

claude-opus-5
the model behind all 4 constrained calls

Messages API
structured output, JSON-schema decoding

classify · propose mentions · select a code index · decode a rule
each call's output space is too small to hold a hallucination

STORAGE

the cache is the version pin a reviewer re-runs against

terms.db · loinc.db
SQLite + FTS5, 232MB terminology cache

semantic/*.npy
precomputed embeddings

cassettes/*.jsonl
recorded model calls
replay with no API key

bundle.*.json / .html
rule sets + reports
one per guideline

cache read / write

compile + execute

CQL ENGINE — JVM + NODE

a second, deterministic runtime — not optional

cql-to-elm
Java · HL7's official CQL → ELM translator

cql-execution
Node · runs ELM against a FHIR bundle

cql-exec-fhir adapts the FHIR data model for cql-execution
ELM only, never CQL — the Java step cannot be skipped

Legend

🟩 Python runtime
🟥 external network service
🟪 JVM / Node — the CQL engine
🟦 storage / local data
⬜ dev & deploy tooling